

Experiment# 06**Implementing the Concept of Friend Functions and Friend Classes****Objective**

Objective of this lab session is to reinforce some basic programming concepts regarding to static members, static member functions, friend functions and friend class.

Software Used

Code Blocks / Dev C++

Theory

A **friend function**, that is a "**friend**" of a given class, is a **function** that is given the same access as methods to private and protected data. A **friend function** cannot be member function of the class. **Friend Class** can access private and protected members of other class in which it is declared as friend. In case of friend class, all of its members are friends. A friend function of a class is defined outside that class' scope but it has the right to access all private and protected members of the class. Even though the prototypes for friend functions appear in the class definition, friends are not member functions.

Example #1

```
#include<iostream>
#include<string.h>
using namespace std;

class data
{
    int num1;
    int num2;
public:
    friend float mean(data);
    void setval(int ,int);
};

void data::setval(int a, int b)
{
    num1=a;
    num2=b;
}
float mean(data d1)
{
    return (d1.num1+d1.num2)/2.0;
}
int main()
```

```

{
data d;
d.setval(5,2);
cout<<"mean:"<<mean(d);

return 0;
}

```

Example #2

```

#include<iostream>
#include<string.h>
using namespace std;

class B;//forward declaration

class A
{
int num1;
public:
void setval(int a)
{
    num1=a;
}
friend void add (A,B);//friend function declaration
};

class B
{
int num2;
public:
void setval(int b)
{
    num2=b;
}
friend void add (A,B);//friend function declaration
};

void add(A CA, B CB)
{
cout<<"\nSum of private data member of two classes are:"<<CA.num1+CB.num2;
}
int main()
{
    A a1;a1.setval(5);
    B b1;b1.setval(3);

```

```

add(a1,b1);
return 0;
}

```

Friend function by value

A value is passed to a function.

Example #3

```

#include<iostream>
#include<string.h>
using namespace std;

```

```

class B;//forward declaration

```

```

class A
{
int num1;
public:
void setval(int a)
{
    num1=a;
}
void display()
{
cout<<"\nClass A value:"<<num1;
}
friend int add(A,B);//friend function declaration
};

```

```

class B
{
int num2;
public:
void setval(int b)
{
    num2=b;
}
void display()
{
cout<<"\nClass B value:"<<num2;
}
friend int add (A,B);//friend function declaration
};

```

```

int add(A CA, B CB)
{
    return CA.num1+CB.num2;
}

```

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```

}
int main()
{
    A a1;a1.setval(5);
    B b1;b1.setval(3);
    a1.display();
    b1.display();
    cout<<"add (a1,b1)";
    return 0;
}

```

Example #4

Friend function by reference

The address of the object is passed to a function.

```

#include<iostream>
#include<string.h>
using namespace std;

class B;//forward declaration

class A
{
    int num1;
public:
    void setval(int a)
    {
        num1=a;
    }
    void display()
    {
        cout<<"\nClass A value:"<<num1;
    }
    friend void Swap(A &,B &);//friend function declaration
};

class B
{
    int num2;
public:
    void setval(int b)
    {
        num2=b;
    }
    void display()
    {
        cout<<"\nClass B value:"<<num2;
    }
}

```

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```

    }
    friend void Swap (A &B, B &); //friend function declaration
};

void Swap(A &CA, B &CB)
{
    int temp=CA.num1;
    CA.num1=CB.num2;
    CB.num2=temp;
}
int main()
{
    A a1;a1.setval(5);
    B b1;b1.setval(3);
    Swap(a1,b1);
    a1.display();
    b1.display();
    return 0;
}

```

Friend Class

In case of friend class, the whole class including its members and functions are friend of that class.

Example #5

Member function of 1 class is friend in another class

```

#include<iostream>
using namespace std;

class B;
class A{
    int a = 2;
public:
    int sub(B &y);
};
class B{
    int b = 11;

public:
    friend int A::sub(B & y);
};

int A::sub(B &y)
{

```

```

        return (a-y.b);
    }

```

```

int main(){
    A a1;
    B b1;
    cout<<a1.sub(b1);
}

```

Example #6

Member function of 1 class is friend in another class (Example 2)

```

#include <iostream>
using namespace std;

class TWO; // forward declaration of class TWO

class ONE{
public:
    void fun(TWO); // member function declaration
};

class TWO{
private:
    int value; // private field
public:
    friend void ONE::fun(TWO); // friend function declaration
};

void ONE::fun(TWO two) // function definition
{
    two.value=10; // accessing private field
    cout<<two.value<<endl;
}

int main(){
    ONE one;
    TWO two;
    one.fun(two);
    return 0;
}

```

Following are some important points about friend functions and classes:

1) Friends should be used only for limited purpose. Too many functions or external classes are declared as friends of a class with protected or private data, it lessens the value of encapsulation of separate classes in object-oriented programming.

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- 2) Friendship is not mutual. If a class A is friend of B, then B doesn't become friend of A automatically.
- 3) Friendship is not inherited

Example #7

```
#include <iostream>
class A {
private:
    int a;
public:
    A() { a=0; } // A() : _a(0){} //constructor
    friend class B; // Friend Class of A};
class B {
private:
    int b;
public:
    void showA(A& x) {
        // Since B is friend of A, it can access private members of A
        cout << "A::a=" << x.a;    }
};
int main() {
    A a;
    B obj;
    obj.showA(a);
    return 0;
}
```

Output:

A::a=0

Lab Tasks

1. Create and implement a program in C++ to define a BOX class and a AREA_VOL class. In the class BOX, data member should be length. Member function should set the value of data member (user defined or by passing argument to the member function). Calculate and display the area and volume of the box in AREA_VOL class using **friend function**. (**Note:** class AREA_VOL should not be friend class of BOX, but only member function/functions of class AREA_VOL can be friend function/functions of class BOX)
2. Create and Implement program in C++ to define 2 classes i.e. A and B. Declare an integer 'a' variables in class A and an integer 'b' in class B. **B class should be friend of class A**. Division (a/b), power (a^b) and multiplication (a*b) of variables a and b must be performed using member function/ functions of class B.

3. Create and Implement program in C++ to define 2 classes i.e. A and B. Declare an integer 'a' variable in class A and an integer 'b' in class B. B and A class should not be friend classes. Display the greater number among 'a' and 'b' using **friend function of class A and B**. (**Note:** friend function should not be member function of either class A or B)